

Maharashtra State Board Of Technical Education, Mumbai																										
Learning and Assessment Scheme Of Diploma Courses for Working Professionals																										
Programme Name		: Diploma In Electronics & Tele-communication Engg. (Working Professional)																								
Programme Code		: PEJ										With Effect From Academic Year					: 2024-25									
Duration of Programme		: 6 Semester																								
Semester		: Third										Scheme					: F									
Sr No	Course Title	Abbreviation	Course Type	Course Code	Diploma Awarding Course	Total IKS Hrs for Sem.	Learning Scheme					Credits	Assessment Scheme													
							Actual Contact Hrs./Week			Self Learning (Activity/ Assignment /Micro Project)	Notional Learning Hrs /Week		Paper Duration (hrs.)	Theory			Based on LL & TL				Based on Self Learning	Total Marks				
							CL	TL	LL					FA-TH	SA-TH	Total	Practical		SLA							
																	FA-PR	SA-PR		Max	Min		Max	Min	Max	Min
																	Max	Max		Max	Min		Max	Min	Max	Min
(All Compulsory)																										
1	DIGITAL TECHNIQUES	DTE	DSC	313303	No	0	3	-	2	1	6	3	3	30	70	100	40	25	10	25#	10	25	10	175		
2	ANALOG ELECTRONICS	ATE	DSC	313324	No	0	3	-	4	1	8	4	3	30	70	100	40	25	10	25#	10	25	10	175		
3	CIRCUITS & NETWORKS	CKN	DSC	313325	No	0	3	-	4	1	8	4	3	30	70	100	40	25	10	25@	10	25	10	175		
4	BASIC PYTHON PROGRAMMING	BPP	AEC	313011	No	0	2	-	2	-	4	2	-	-	-	-	-	25	10	25@	10	-	-	50		
5	ESSENCE OF INDIAN CONSTITUTION	EIC	VEC	313002	No	0	1			1	2	1	-	-	-	-	-	-	-	-	-	50	20	50		
Total						0	12	0	12	4		14		90	210	300		100		100		125		625		
Abbreviations : CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, FA - Formative Assessment,SA -Summative Assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment Legends : @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination Note : 1. FA-TH represents average of two class tests of 30 marks each conducted during the semester. 2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester. 3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work. 4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks 5. 1 credit is equivalent to 30 Notional hrs. 6. * Self learning hours shall not be reflected in the Time Table. 7. * Self learning includes micro project / assignment / other activities. Course Category : Discipline Specific Course Core (DSC) , Discipline Specific Elective (DSE) , Value Education Course (VEC) , Intern./Apprenti./Project./Community (INP) , AbilityEnhancement Course (AEC) , Skill Enhancement Course (SEC) , GenericElective (GE)																										

DIGITAL TECHNIQUES**Course Code : 313303**

Programme Name/s	: Artificial Intelligence/ Artificial Intelligence and Machine Learning/ Automation and Robotics/ Computer Technology/ Computer Engineering/ Computer Science & Engineering/ Digital Electronics/ Data Sciences/ Electronics & Tele-communication Engg./ Electrical and Electronics Engineering/ Electronics & Communication Engg./ Electronics Engineering/ Computer Hardware & Maintenance/ Instrumentation & Control/ Industrial Electronics/ Instrumentation/ Medical Electronics/ Computer Science/ Electronics & Computer Engg.
Programme Code	: AI/ AN/ AO/ CM/ CO/ CW/ DE/ DS/ EJ/ EK/ ET/ EX/ HA/ IC/ IE/ IS/ MU/ SE/ TE
Semester	: Third
Course Title	: DIGITAL TECHNIQUES
Course Code	: 313303

I. RATIONALE

Digitization implies use of digital circuits in most of automation and industrial systems. The knowledge of logic gates, combinational and sequential circuits using discrete gates and digital ICs will enable students to interpret working of digital equipment and test their functionality.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to help students to attain the following industry/employer expected outcome through various teaching learning experiences:

Student will able to test the functionality of the digital circuits/system.

III. COURSE LEVEL LEARNING OUTCOMES (COS)**MSBTE Approval Dt. 02/07/2024****Semester - 3, K Scheme**

DIGITAL TECHNIQUES**Course Code : 313303**

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Apply number system and codes concept to interpret working of digital systems.
- CO2 - Apply Boolean laws to minimize complex Boolean function.
- CO3 - Develop combinational logic circuits for given applications.
- CO4 - Develop sequential logic circuits using Flip-flops.
- CO5 - Interpret the functions of data converters and memories in digital electronic systems.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL	FA-TH				SA-TH		Total		FA-PR		SA-PR		SLA		
Max	Max	Max	Min	Max	Min	Max	Min	Max	Min												
313303	DIGITAL TECHNIQUES	DTE	DSC	3	-	2	1	6	3	3	30	70	100	40	25	10	25#	10	25	10	175

DIGITAL TECHNIQUES**Course Code : 313303****Total IKS Hrs for Sem. : 0 Hrs**

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's)aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
-------	---	--	--------------------------------

DIGITAL TECHNIQUES**Course Code : 313303**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Convert the given number from one number system to another number system. TLO 1.2 Perform arithmetic operations on binary numbers. TLO 1.3 Subtract given binary numbers using 1's and 2's complement method. TLO 1.4 Convert the given coded number into the other specified code. TLO 1.5 Write the application of the given code. TLO 1.6 Perform BCD addition and subtraction for the given Decimal numbers .	Unit - I Number Systems 1.1 Number Systems: Types of Number Systems (Binary, Octal, Decimal, Hexadecimal), conversion of number systems 1.2 Binary Arithmetic: Addition, Subtraction, Multiplication and Division 1.3 Subtraction using 1's and 2's complement method 1.4 Codes: BCD, Gray code, Excess-3 and ASCII code, Code conversions, Applications of codes. 1.5 BCD Arithmetic: BCD Addition, Subtraction using 9's and 10's complement	Lecture Using Chalk-Board

DIGITAL TECHNIQUES**Course Code : 313303**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Define the given characteristics parameters of the digital logic families. TLO 2.2 Draw symbol and truth table of given logic gates. TLO 2.3 Explain the concept of Buffer and Tristate logic . TLO 2.4 Implement basic gates and other gates with the help of universal gate. TLO 2.5 Simplify the given expression using Boolean laws and develop logic circuits .	Unit - II Logic Gates and Boolean Algebra 2.1 Logic Families: Characteristics Parameters of logic Families- Noise margin, Power dissipation, Figure of merit ,Fan in, Fan out, Speed of operation, maximum clock frequency supply voltage requirement ,power per gate , Comparison of TTL, CMOS and ECL logic family 2.2 Introduction to positive and negative logic systems, Logic Gates: Symbol ,Truth table of Basic logic gates(AND,OR,NOT),Universal gates(NAND,NOR) and Special purpose gates(EX-OR,EX-NOR) 2.3 Buffer: Tristate logic, Unidirectional and Bidirectional 2.4 Boolean algebra : Laws of Boolean algebra, Duality Theorem ,De-Morgan's theorem	Flipped Classroom Lecture Using Chalk-Board

DIGITAL TECHNIQUES**Course Code : 313303**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Develop logic circuits for standard SOP/POS form of the given logic expression. TLO 3.2 Minimize the given logic expression using K-map (up to 4 variables). TLO 3.3 Design Adder and subtractor using K-map. TLO 3.4 Describe working of specified Encoder and Decoder with help of block diagram and truth table. TLO 3.5 Describe the working of Multiplexer and Demultiplexer.	Unit - III Combinational Logic Circuits 3.1 Standard Boolean expression: Sum of products [SOP] and Products of Sum [POS], Min-term and Max-term, SOP-POS form conversion, realisation using NAND/NOR gates 3.2 Boolean Expression reduction using K-map: Minimization of Boolean expressions (upto 4 variables) using SOP and POS form 3.3 Arithmetic circuits : design Half and Full Adder using K-maps, design Half and Full Subtractor using K-maps , n bit adder and n bit subtractor . 3.4 Encoder and Decoder: Functions of Encoder and Decoder, Block Diagram and Truth table, Priority Encoder (4:2, 8:3), BCD to 7 segment Decoder/Driver, Keyboard Encoder / decoder 3.5 Multiplexer and Demultiplexer: Working, Truth table and applications of MUX and DEMUX. MUX tree, DEMUX tree, DEMUX as Decoder	Flipped Classroom Presentations Lecture Using Chalk-Board

DIGITAL TECHNIQUES**Course Code : 313303**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Differentiate between Latch and Flip Flop. TLO 4.2 Explain basic memory cell and use relevant triggering technique for the given digital circuit. TLO 4.3 Describe the truth tables for the given Flip flops, applications of Flip flops. TLO 4.4 Use the given type of flip flop and its excitation table to design specific type of counter. TLO 4.5 Describe the working of specified shift register with the help of timing diagram. TLO 4.6 Design specified modulo-N counter using Flip flops . TLO 4.7 Design Ring	Unit - IV Sequential Logic Circuits 4.1 Difference between Combinational and Sequential Logic circuits, Time independent (un-clocked)and Time dependent (Clocked) logic system , Flips- Flops and Latch, Basic memory cell ,RS-Latch using NAND and NOR, Triggering methods- Edge trigger and Level Trigger 4.2 Flip-Flops: S-R, J-K, T and D, Truth table and logic circuits of each flip-flop, Excitation table, applications 4.3 Race around condition in JK flip-flop, Master- Slave JK Flip Flop 4.4 Shift registers- Serial In Serial Out, Serial In Parallel Out, Parallel In Serial Out ,Parallel In Parallel Out,Bi-directional Shift register, 4-bit Universal Shift register 4.5 Counters- Synchronous and Asynchronous counters, Modulus of counter, Ripple counter, Ring Counter, Twisted Ring Counter, Up – down counter, Decade Counter, MOD-N counter, Timing Diagram	Video Demonstrations Lecture Using Chalk-Board Simulation

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

DIGITAL TECHNIQUES**Course Code : 313303**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
	/Twisted ring counter using given Flip-Flop.		
5	TLO 5.1 Describe the working of the given type of DAC. TLO 5.2 Calculate the output voltage for the given digital input for specified DAC. TLO 5.3 Describe the working of the given type of ADC. TLO 5.4 Compare working of ROM, EPROM, EEPROM and Flash Memory .	Unit - V Data Converters and Memories 5.1 Digital to Analog Data Converter (DAC)- circuit diagram and working of Weighted resistor DAC and R-2R Ladder DAC, DAC Specification/Selection factors 5.2 Analog to Digital Data Converter (ADC) : Block Diagram, Types and Working of Dual Slope ADC, Successive Approximation, Flash Type ADC, ADC selection factors/ specifications 5.3 Memories: Types- Primary memory , Secondary Memory, Organization, Dimension, Memory Bank, Features , Applications: RAM (SRAM, DRAM), Volatile and Non-Volatile, ROM (PROM, EPROM, EEPROM), Flash Memory, Comparison of RAM and ROM, EPROM and Flash Memory, SIMM: Features, SSD memory: Features,	Video Demonstrations Lecture Using Chalk-Board

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
---	--------------	---	-----------------------	---------------------

DIGITAL TECHNIQUES**Course Code : 313303**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Test the functionality of basic gates. LLO 1.2 Test the functionality of special purpose gates.	1	* Test the functionality of AND, OR, NOT, Ex-OR and EX-NOR logic Gates using equivalent 74 series or CMOS Devices [CD] series.	2	CO1 CO2
LLO 2.1 Test the functionality of NAND and NOR gate using breadboard.	2	* Test the functionality of the given Universal Gates using equivalent 74 series /CD series.	2	CO2
LLO 3.1 Test the functionality of the constructed Basic gates using universal gates.	3	* Construct Basic Gates using Universal Gates.	2	CO2
LLO 4.1 Construct Ex-OR, EX- NOR gates using universal gates.	4	Construct Exclusive Gates using Universal Gates.	2	CO2
LLO 5.1 Build the logic circuit on breadboard to verify the De -Morgan's theorems.	5	* Verify De-Morgan's Theorem (1 and 2).	2	CO2
LLO 6.1 Verify the truth table of Half and Full adder circuits for the given input.	6	* Implement 2 input, 3 input Adder Circuit.	2	CO3
LLO 7.1 Verify the truth table of Half and Full subtractor using Boolean expressions.	7	Implement 2 input, 3 input Subtractor Circuit.	2	CO3
LLO 8.1 Construct and test BCD to 7 segment decoder using Digital IC.	8	Test the output of BCD to 7 Segment Decoder using Digital IC for the given inputs.	2	CO3

DIGITAL TECHNIQUES**Course Code : 313303**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 9.1 Build/Test 2 or 4 bit Magnitude comparator using Digital IC.	9	Check the output of comparator circuit consisting of Digital IC.	2	CO3
LLO 10.1 Build / test function of MUX Digital IC.	10	* Build and test the functionality of 4:1/8:1 Multiplexer.	2	CO3
LLO 11.1 Build / test function of DEMUX Digital IC.	11	Build and test the functionality of 1:4/1:8 De-Multiplexer.	2	CO3
LLO 12.1 Test functionality of RS flip flop using NAND Gate .	12	Implement and verify the truth table of RS Flip flop.	2	CO4
LLO 13.1 Test functionality of Master Slave (MS) JK flip-flop using Digital IC.	13	Implement and test the functionality of master slave-JK Flip Flop using Digital IC.	2	CO4
LLO 14.1 Test functionality and truth table for D and T Flip flop.	14	Use Digital IC to construct and test the functionality of D and T flip flop.	2	CO4
LLO 15.1 Interpret timing diagram of 4 bit Universal Shift Register.	15	Build 4- bit Universal Shift register and observe the timing diagram.	2	CO4
LLO 16.1 Interpret timing diagram of 4-bit ripple counter using Digital IC.	16	Implement Ripple Counter using Digital IC.	2	CO4
LLO 17.1 Interpret timing diagram of Decade counter (Mod-10).	17	* Implement Decade Counter Using Digital IC.	2	CO4
LLO 18.1 Build R-2R resistive network on breadboard to convert given digital data into analog.	18	* Test the output of given R-2R type Digital to Analog Converter for the given input.	2	CO5

DIGITAL TECHNIQUES**Course Code : 313303**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- Implement 1:8 DEMUX using 1:4 /1:2 DE-MUX.
- Build a circuit to implement 4 Bit adder.
- Build a 4bit parity generator and parity tester.
- Implement 16:1 MUX using 8:1/4:1 MUX.
- Build a circuit to test 7 bit segment display.
- Build a LED display bar.
- Develop a project on Burglar alarm.
- Light Detector circuit using NAND gate.

DIGITAL TECHNIQUES**Course Code : 313303****Note :**

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Digital Storage Oscilloscope 25MHz/60MHz/70MHz/100MHz Dual Channel, 4 Trace CRT / TFT based X10 magnification 20 nS max sweep rate, Alternate triggering Component tester and with optional features such as Digital Read out, USB interface. Any other Oscilloscope with additional features is also suitable with magnifying probe at least two probes, if possible isolated probe	15,16,17
2	Trainer kit for 4 bit Counter using Flip Flops 4 bit ripple counter synchronous counter IC 7476 based circuit, Input given by switches and output indicated on LED, Facility to select MOD 8 or MOD 16 mode, Built in DC power supply and manual pulser with indicator	16,17

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

DIGITAL TECHNIQUES**Course Code : 313303**

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
3	Trainer kit IC DAC IC 0800 Trainer based on IC 0800, 8 bit digital input selected by switches and provision for measurement of analog output. Facility to study effect of change in reference voltage, Built in buffer amplifier, Built in DC power supply	18
4	Digital multimeter 3.5 digit with R , V, I measurements, diode and BJT testing	All
5	Digital IC Tester Tests a wide range of Analog and Digital ICs such as 74 series /CD series	All
6	Bread Board Development System Bread Board system with DC power output 5V,+/-12V and 0-5V variable , digital voltmeter ,ammeter , LED indicators 8 no , logic input switches 8 no, 7 segment display 2 no, clockgenerator	All
7	Trainer kits for digital ICs Trainer kit should consists of digital ICs for logic gates, flop flop, shift registers, counter alongwith toggle switches for inputs and bi-colourLED at outputs, built in power supply	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Number Systems	CO1	5	2	4	2	8
2	II	Logic Gates and Boolean Algebra	CO2	8	2	4	6	12
3	III	Combinational Logic Circuits	CO3	12	4	6	8	18
4	IV	Sequential Logic Circuits	CO4	12	4	6	8	18
5	V	Data Converters and Memories	CO5	8	4	6	4	14
Grand Total				45	16	26	28	70

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

DIGITAL TECHNIQUES**Course Code : 313303****X. ASSESSMENT METHODOLOGIES/TOOLS****Formative assessment (Assessment for Learning)**

- Two offline unit tests of 30 marks and average of two unit test marks will be consider for out of 30 marks.
- Each practical will be assessed considering 60% weightage to process, 40% weightage to product.
- For formative assessment of laboratory learning 25 marks

Summative Assessment (Assessment of Learning)

- End semester assessment is of 70 marks.
- End semester summative assessment of 25 marks for laboratory learning

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	-	1	-	-	-	3			

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

DIGITAL TECHNIQUES**Course Code : 313303**

CO2	2	-	2	-	-	-	2			
CO3	3	2	3	2	-	1	2			
CO4	3	2	3	2	-	1	2			
CO5	2	-	2	2	1	1	2			

Legends :- High:03, Medium:02,Low:01, No Mapping: -

*PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Jain R.P	Modern Digital Electronics	McGraw-Hill Publishing, New Delhi,2009 ISBN:9780070669116
2	Anand Kumar	Fundamentals of Digital Circuits	PHI learning Private limited, ISBN:978-81-203-5268-1
3	Salivahanan S, Arivazhagan S.	Digital Circuits and Design	Vikas Publishing House, New Delhi,2013 ISBN: 9789325960411
4	Puri.V.K	Digital Electronics	McGraw-Hill Publishing, New Delhi,2016 ISBN:97800746331751
5	Malvino A.P Donald .P. Leach	Digital Principles	McGraw-Hill Education, New Delhi ISBN:9789339203405
6	Anil.K.Maini	Digital Electronics: Principles, Devices and Applications	Wiley India, Delhi, 2007, ISBN:9780470032145
7	Floyd, Thomas	Digital Fundamentals	Pearson Education India, Delhi 2014,ISBN:9780132737968
8	G.K.Kharate	Digital Electronics	Publisher: Oxford University Press, ISBN: 9780198061830

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

DIGITAL TECHNIQUES**Course Code : 313303****XIII . LEARNING WEBSITES & PORTALS**

Sr.No	Link / Portal	Description
1	https://studytronics.weebly.com/digital-electronics.html	Basics of Digital Electronics
2	https://www.udemy.com/course/basics-of-digital-techniques/	Introduction To Digital Number System & Logic Gates
3	https://www.geeksforgeeks.org/synchronous-sequential-circuits-in-digital-logic/	Boolean Algebra and Logic Gates, Combinational and Sequential Logic Circuits
4	https://onlinecourses.nptel.ac.in/noc19_ee51/preview	Digital Circuits
5	https://de-iitr.vlabs.ac.in/	Virtual Labs for Digital Systems
6	https://www.tutorialspoint.com/digital_circuits/digital_circuits_sequential_circuits.htm	Sequential Circuits
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ANALOG ELECTRONICS

Course Code : 313324

Programme Name/s	: Digital Electronics/ Electronics & Tele-communication Engg./ Electrical and Electronics Engineering/ Electronics & Communication Engg./ Electronics Engineering/ Instrumentation & Control/ Industrial Electronics/ Instrumentation/ Medical Electronics/ Electronics & Computer Engg.
Programme Code	: DE/ EJ/ EK/ ET/ EX/ IC/ IE/ IS/ MU/ TE
Semester	: Third
Course Title	: ANALOG ELECTRONICS
Course Code	: 313324

I. RATIONALE

Analog electronic circuits are the basic building blocks of many complex electronic system. Therefore it is necessary for students to understand the working principle and testing of basic analog circuits consist of discrete components and integrated circuits. After learning this course students will be able to apply the concept of working of basic electronic circuit and Op-Amp circuits to maintain the electronic system.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to attain the following industry/ employer expected outcome through various teaching learning experiences:

Maintain analog electronic circuits.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Use transistor as a Power Amplifier.

ANALOG ELECTRONICS**Course Code : 313324**

- CO2 - Construct various configurations of Op-Amp for different applications.
- CO3 - Maintain different waveform generator circuits.
- CO4 - Analyze active filters used in various electronic circuits.
- CO5 - Use specific analog IC to develop various applications.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL	FA-TH	SA-TH			Total		FA-PR		SA-PR		SLA				
Max	Max	Max	Min	Max	Min	Max	Min	Max	Min												
313324	ANALOG ELECTRONICS	ATE	DSC	3	-	4	1	8	4	3	30	70	100	40	25	10	25#	10	25	10	175

ANALOG ELECTRONICS**Course Code : 313324****Total IKS Hrs for Sem. : 0 Hrs**

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
-------	---	---	--------------------------------

ANALOG ELECTRONICS**Course Code : 313324**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Describe the performance of the given power amplifier parameters. TLO 1.2 Explain with sketches the working of given type of power amplifier. TLO 1.3 Compare the given type of power amplifiers on the basis of performance parameter. TLO 1.4 Select relevant type of power amplifier for the given applications.	Unit - I Power Amplifiers 1.1 Power Amplifier: Concept, Performance parameters like: Gain, Bandwidth, frequency band, efficiency and distortion 1.2 Classification: Class A, Class B, Class AB and Class C power amplifier and their applications 1.3 Circuit diagram, working, input output waveforms and efficiency of single stage Class A, Class B, Class AB and Class C power amplifier, Push Pull amplifier, Complementary symmetry push-pull amplifier. Transformer less push-pull amplifier. Distortions in Power amplifier 1.4 Generalised features of audio power amplifier IC's, Heat Sink	Lecture Using Chalk-Board Video Demonstrations Assignments

ANALOG ELECTRONICS**Course Code : 313324**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Describe with sketches the function of the given block(s) of Op-Amp. TLO 2.2 List ideal characteristics of Op-Amp. TLO 2.3 Define the given parameters of Op-Amp. TLO 2.4 Interpret the output waveform of given mode of operation of Op-Amp. TLO 2.5 Calculate the output voltage of given arithmetic circuit consist of Op-Amp. TLO 2.6 Describe the working of the given application circuit consist of Op-Amp.	Unit - II Op-Amp and its applications 2.1 Op-Amp: block diagram, symbol, characteristics, open loop and closed loop amplifier, virtual ground concept, IC-741 and it's pin configuration 2.2 Op-Amp parameters: input offset voltage, output offset voltage, input offset current, input bias current, Common Mode Rejection Ratio, Power supply rejection ratio ,slew rate, input and output Impedance, bandwidth and gain bandwidth product, Drift parameters 2.3 Closed loop configurations: inverting and non-inverting 2.4 Basic mathematical applications such as adder, subtractor, integrator and differentiator 2.5 Sample and Hold circuit, I-V converter, V-I converter 2.6 Comparator: Zero Crossing Detector (Inverting and Non Inverting type), Schmitt Trigger, Window Detector and Peak Detector	Lecture Using Chalk-Board Video Demonstrations Assignments

ANALOG ELECTRONICS**Course Code : 313324**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Explain the principle of positive and negative feedback for amplifier circuit. TLO 3.2 State Barkhausen's criteria for oscillation. TLO 3.3 Describe the working principle of given type of oscillator. TLO 3.4 Calculate the frequency of oscillation for given oscillator. TLO 3.5 Select the circuit components of the given type of oscillator for given operating frequency.	Unit - III Waveform Generators 3.1 Principle of feedback amplifier. Types of feedback: Negative and Positive feedback, advantages and disadvantages of negative feedback, Types of feedback connections (Block Diagrams and features only) 3.2 Oscillator: need of oscillator, compare oscillator and amplifier 3.3 Condition for oscillations: Barkhausen's criteria, classification of oscillator 3.4 Oscillators Circuits : Phase shift oscillator using IC 741 , Hartley oscillator using IC 741 and crystal oscillator using BJT & FET	Lecture Using Chalk-Board Video Demonstrations Assignments Blended Classroom

ANALOG ELECTRONICS**Course Code : 313324**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Describe with circuit diagram working of the given type of filter. TLO 4.2 Identify the type of filter based on given frequency response. TLO 4.3 Calculate the cut off frequency of given type of filter. TLO 4.4 Develop given type and order of active filter for the given cut off frequency.	Unit - IV Active Filters 4.1 Definition, type of filters and difference between active and passive filter 4.2 Merits and demerits of active filter over passive filter 4.3 Terms related to filters: Order of filter, cut off frequency, Pass band, Stop band, Center frequency, Roll off rate, Bandwidth and Q factor 4.4 Order and frequency response: First and second order of low pass and high pass filter	Lecture Using Chalk-Board Assignments Video Demonstrations Blended Learning Tools
5	TLO 5.1 Develop a circuit for the given application using IC-555. TLO 5.2 Calculate the duty cycle of the given type of multivibrator. TLO 5.3 Describe the working of given block of PLL. TLO 5.4 Develop a circuit for the given application using IC-565.	Unit - V Specialized IC Applications 5.1 Timer IC: IC 555 block diagram, pin diagram and functions, Astable and Monostable multivibrators, Voltage Controlled Oscillator 5.2 Phase Lock Loop (PLL) : IC 565 pin diagram, block diagram and it's working, Lock range and Capture range 5.3 Applications of PLL: PLL as a frequency multiplier, FM Demodulator	Lecture Using Chalk-Board Assignments Video Demonstrations Blended Learning Tools

ANALOG ELECTRONICS**Course Code : 313324****VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Interpret the operation of single stage Class A power amplifier.	1	* Test the performance of single stage Class A power amplifier.	2	CO1
LLO 2.1 Interpret the operation of Class B push pull power amplifier.	2	* Test the performance of Class B push pull power amplifier.	2	CO1
LLO 3.1 Interpret the operation of Class AB power amplifier.	3	Test the performance of Class AB power amplifier.	2	CO1
LLO 4.1 Measure output voltage swing of Op-Amp (IC 741).	4	* Determine the range of output voltage swing of Op-Amp (IC 741).	2	CO2
LLO 5.1 Measure input offset voltage and output offset voltage of IC 741.	5	* Build the circuit to measure input offset voltage and output offset voltage of IC 741.	2	CO2
LLO 6.1 Connect IC 741 in inverting and non-inverting mode. LLO 6.2 Measure the voltage gain of inverting and non-inverting amplifier circuit using IC 741.	6	* Determine the gain of inverting and non-inverting amplifier using IC 741.	2	CO2
LLO 7.1 Select the proper range of multimeter to measure the voltage. LLO 7.2 Measure output voltage of adder circuit consist of IC 741.	7	* Build /Test adder circuit consist of IC 741.	2	CO2

ANALOG ELECTRONICS**Course Code : 313324**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 8.1 Select the proper range of multimeter to measure the voltage. LLO 8.2 Measure output voltage of subtractor circuit consist of IC 741.	8	Build /Test subtractor circuit consist of IC 741.	2	CO2
LLO 9.1 Use function generator. LLO 9.2 Interpret input and output waveforms of Integrator circuit consist of IC 741.	9	* Build /Test Integrator circuit consist of IC 741.	2	CO2
LLO 10.1 Use function generator. LLO 10.2 Interpret input and output waveforms of Differentiator circuit consist of IC 741.	10	* Build /Test Differentiator circuit consist of IC 741.	2	CO2
LLO 11.1 Measure output current of V to I converter circuit using IC 741.	11	* Build/ Test V to I converter circuit using IC 741.	2	CO2
LLO 12.1 Interpret output voltage waveform of zero crossing detector.	12	* Build the circuit of zero crossing detector and test the output.	2	CO2
LLO 13.1 Check the performance of feedback on the output voltage of amplifier.	13	Use transistor to build/test voltage series feedback amplifier with feedback.	2	CO3
LLO 14.1 Check the performance of feedback on the output voltage of amplifier.	14	Use transistor to build/test voltage shunt feedback amplifier with feedback.	2	CO3
LLO 15.1 Check the performance of amplifier for positive and negative feedback.	15	* Test the effect of positive and negative feedback on the output voltage of given amplifier.	2	CO3

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ANALOG ELECTRONICS**Course Code : 313324**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 16.1 Measure the output frequency of RC phase shift oscillator based on IC 741.	16	* Test the circuit to measure the frequency of oscillation of the given RC phase shift oscillator consist of IC 741.	2	CO3
LLO 17.1 Measure the output frequency of Crystal Oscillator	17	Test the circuit of transistorised Crystal Oscillator	2	CO3
LLO 18.1 Measure the output frequency of Hartley Oscillator consist of IC 741.	18	Test the Hartley Oscillator based on IC 741.	2	CO3
LLO 19.1 Observe the output waveform of Hartley Oscillator.	19	Simulate the working of Hartley Oscillator using multisim or relevant software.	2	CO3
LLO 20.1 Measure bandwidth and cut off frequency of low pass filter. LLO 20.2 Plot the frequency response of low pass filter.	20	* Build and test the circuit of first order low pass filter.	2	CO4
LLO 21.1 Measure bandwidth and cut off frequency of high pass filter. LLO 21.2 Plot the frequency response of high pass filter.	21	* Build and test the circuit of first order high pass filter.	2	CO4
LLO 22.1 Observe the output waveform of high pass filter.	22	Simulate the working of high pass filter consist of IC 741 using multisim or relevant software.	2	CO4
LLO 23.1 Select the proper value of R and C for generating waveform of specific duty cycle.	23	* Build / test astable multivibrator using IC 555 for the specific duty cycle.	2	CO5
LLO 24.1 Measure the time period of monostable multivibrator using IC 555.	24	Build / test monostable multivibrator using IC 555 for the specific duty cycle.	2	CO5

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ANALOG ELECTRONICS**Course Code : 313324**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 25.1 Observe the output waveform of monostable multivibrator.	25	Simulate the working of monostable multivibrator using IC 555 using multisim or relevant software.	2	CO5
LLO 26.1 Check the performance of Voltage Controlled Oscillator using IC 555.	26	* Build/ Test Voltage Controlled Oscillator using IC 555.	2	CO5
LLO 27.1 Measure the output frequency of multiplier circuit consist of IC565.	27	Build/ test the circuit of frequency multiplier using PLL IC 565.	2	CO5
LLO 28.1 Interpret input and output waveform of FM demodulator circuit contains PLL (IC 565).	28	Check the performance of PLL as FM demodulator (IC 565).	2	CO5
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- 1) Construct audio amplifier using IC LM386N or equivalent IC.
- 2) Develop clap switch using IC 741.
- 3) Build automatic light operated switch using LDR and IC 741.
- 4) Build automatic evening lamp using IC 555 or equivalent IC.

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ANALOG ELECTRONICS**Course Code : 313324**

- 5) Construct square wave generator using IC 741 or equivalent.
- 6) Develop low pass filter/high pass filter with cut off frequency of 2KHz using universal IC UAF42 or equivalent IC.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Cathode Ray Oscilloscope Dual Trace 20Mhz/30Mhz, 1Mega ohm Input Impedance.	1,2,3,9,10,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26
2	Digital Storage Oscilloscope 20MHz and above, 1Mega ohm Input Impedance.	1,2,3,9,10,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26
3	Function Generator 0-2 MHz with sine, square and triangular output with variable amplitude and frequency.	1,2,3,9,10,12,13,14,15,19,20,21,22,23,24,25,26
4	Simulation software like: TINA-TI/Multisim etc. or any other open-source simulation software can be used.	19,22,25

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ANALOG ELECTRONICS**Course Code : 313324**

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
5	Dual Tracking Power Supply 0-30V,2 A, short circuit and over voltage protection.	3,4,5,6,7,8,9,10,11,16,17,18,19,20,21
6	Analog IC tester with tests Op-amp,555, IC testing 741 , 555 ,556 ,565 it has Auto search facility of IC's Test by: Truth table/sequence table comparison ZIF: 40 pin DIP ZIF sockets.	4,5,6,7,8,9,10,11,12,16,17,18,19,20,21,22,23,24,25,26
7	Variable DC power supply 0-30V, 2A with short circuit and over voltage protection.	All
8	Digital Multimeter: Minimum 3 ½ digit 4 ½ digit display,9999 counts digital multimeter measures Vac, Vdc(1000V max) , Adc, Aac (10 amp max), Resistance(0-100 M ohm), diode and transistor testing mode.	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Power Amplifiers	CO1	8	4	4	4	12
2	II	Op-Amp and its applications	CO2	11	4	6	8	18
3	III	Waveform Generators	CO3	8	4	4	6	14
4	IV	Active Filters	CO4	8	4	4	6	14
5	V	Specialized IC Applications	CO5	10	4	4	4	12
Grand Total				45	20	22	28	70

X. ASSESSMENT METHODOLOGIES/TOOLS**MSBTE Approval Dt. 02/07/2024****Semester - 3, K Scheme**

ANALOG ELECTRONICS**Course Code : 313324****Formative assessment (Assessment for Learning)**

- Two offline unit tests of 30 marks and average of two unit test marks will be consider for out of 30 marks.
- For formative assessment of laboratory learning 25 marks.
- Each practical will be assessed considering 60% weightage to process, 40% weightage to product.

Summative Assessment (Assessment of Learning)

- End semester assessment is of 70 marks.
- End semester summative assessment is of 25 marks for laboratory learning.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	3	1	2	2	1	--	2			
CO2	3	2	2	2	1	--	2			
CO3	2	2	2	2	1	1	2			

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ANALOG ELECTRONICS**Course Code : 313324**

CO4	2	3	2	1	1	1	1			
CO5	2	3	2	2	1	1	2			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
 *PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Sedha R.S.	Applied Electronics	S.Chand, New Delhi,2015 ISBN:9788121927833
2	Gayakwad Ramakant A.	Op-Amps and Linear Integrated Circuits	PHI Learning, New Delhi 2011, ISBN:9788120320581
3	Salivahanan S., Bhaaskaran Kanchana V. S.	Linear Integrated Circuits	Tata McGraw-Hill Education,New Delhi, 2018, ISBN: 9789353160487
4	Chaudhary D. Roy	Linear Integrated Circuits	New Age International Publishers ISBN: 9788122420906
5	S. Salivahanan, N. Suresh Kumar	Electronic Devices and Circuits	McGraw Hill Education, Edition-Fourth, ISBN-9789339219505

XIII. LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.youtube.com/watch?v=mgoCeOCjiBI	Basic of Op-Amp.
2	https://youtu.be/dKTbrZMscpM?si=6lQ3xdhGvLDZL-VL	Class A Power Amplifier
3	https://www.youtube.com/watch?v=BfjdB09V1NQ&list=PLuv3GM6-gsE3npYPJJDnEF3pdiHJT6Kj3&index=22	Op-Amp as Integrator
4	https://www.youtube.com/watch?v=M3yI0byaqKc&list=PLuv3GM6-gsE3npYPJJDnEF3pdiHJT6Kj3&index=27	Introduction to Oscillator.

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ANALOG ELECTRONICS**Course Code : 313324**

Sr.No	Link / Portal	Description
5	https://www.youtube.com/watch?v=aeQoEnH74C8&list=PLuv3GM6-gsE3npYPJJDnEF3pdiHZT6Kj3&index=31	Working of Crystal Oscillator.
6	https://www.elprocus.com/op-amp-applications-in-electronics/	Applications of Op-Amp.
7	https://testbook.com/electrical-engineering/power-amplifier-definition-types-and-uses	Power Amplifiers
8	http://vlabs.iitkgp.ac.in/be/exp17/inverting_opamp.html	Virtual Lab for Inverting Amplifier
9	https://ae-iitr.vlabs.ac.in/exp/astable-monostable-multivibrator/simulation/astable/exp.html?	Virtual Lab for Multivibrators using Op-Amp.
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

CIRCUITS & NETWORKS

Course Code : 313325

Programme Name/s	: Digital Electronics/ Electronics & Tele-communication Engg./ Electrical and Electronics Engineering/ Electronics & Communication Engg./ Electronics Engineering/ Industrial Electronics
Programme Code	: DE/ EJ/ EK/ ET/ EX/ IE
Semester	: Third
Course Title	: CIRCUITS & NETWORKS
Course Code	: 313325

I. RATIONALE

Diploma engineers must deal with the electronic circuit while maintaining various electronic equipment/systems in the industry. This course will help the students to use principles of circuit and analyse to maintain the electric circuit/network.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to help the student to attain the following industry/ employer expected outcome through various teaching learning experiences

Measure and interpret Electric circuits/networks parameters.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - Calculate voltage and current of the given circuit using nodal and mesh analysis.
- CO2 - Use various network theorems to calculate circuit parameters.
- CO3 - Determine the circuit parameters of two port network.
- CO4 - Calculate the electrical parameters of single phase A.C. circuit.
- CO5 - Find the resonance condition of electric/electronic circuits.

CIRCUITS & NETWORKS**Course Code : 313325****IV. TEACHING-LEARNING & ASSESSMENT SCHEME**

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL	FA-TH	SA-TH			Total		FA-PR		SA-PR		SLA				
Max	Max	Max	Min	Max	Min	Max	Min	Max	Min												
313325	CIRCUITS & NETWORKS	CKN	DSC	3	-	4	1	8	4	3	30	70	100	40	25	10	25@	10	25	10	175

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

CIRCUITS & NETWORKS**Course Code : 313325****V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Define various terms related to circuit and network. TLO 1.2 Use KCL and KVL to calculate voltage and current in the given resistive circuit. TLO 1.3 Apply nodal analysis to find voltage and current of the given network. TLO 1.4 Apply mesh analysis to determine voltage and current of the given network. TLO 1.5 Convert the given star connection to delta connection and vice versa. TLO 1.6 Use source conversion techniques to simplify the given circuit. TLO 1.7 Analyze the transient response of RL, RC and RLC series circuit.	Unit - I DC Network Analysis 1.1 Terms related to circuit and network: Node, Branch, Loop, Mesh 1.2 Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL) 1.3 Nodal Analysis 1.4 Mesh Analysis 1.5 Star-delta and delta- star conversion 1.6 Conversion of voltage to current source and current to voltage source 1.7 Transient Response : RL series circuit, RC series circuit , RLC Series circuit	Lecture Using Chalk-Board Video Demonstrations Assignments

CIRCUITS & NETWORKS**Course Code : 313325**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
2	TLO 2.1 Use superposition theorem to calculate voltage and current in the given multisource circuit. TLO 2.2 Use Thevenin's theorem to simplify the complex circuit. TLO 2.3 Use Norton's theorem to simplify the complex circuit. TLO 2.4 Calculate load impedance using maximum power transfer theorem for the given circuit. TLO 2.5 Use reciprocity theorem to analyse the given circuit.	Unit - II Network Theorems 2.1 Superposition theorem for both AC voltage and DC source 2.2 Thevenin's theorem 2.3 Norton's theorem 2.4 Maximum power transfer theorem 2.5 Reciprocity theorem	Lecture Using Chalk-Board Video Demonstrations Blended learning Assignments

CIRCUITS & NETWORKS**Course Code : 313325**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Differentiate the given types of networks. TLO 3.2 Calculate Z, Y parameters of the given circuit. TLO 3.3 Find the ABCD parameters of the given circuit. TLO 3.4 Connect two port networks in series, parallel and cascade configuration and analyze the output. TLO 3.5 List the types and applications of attenuator. TLO 3.6 List the types and applications of passive filters.	Unit - III Analysis of two port network 3.1 Network Types: Active and Passive, Bilateral and Unilateral, Linear and Nonlinear, Symmetrical and Asymmetrical, Single port and Two port network 3.2 Open circuit(Z) and short circuit(Y) parameters 3.3 Transmission (ABCD) parameters 3.4 Interconnection of two port network- series, parallel and cascade configuration 3.5 Attenuators: Definition, types-T and Pi, features, frequency response, applications, comparison 3.6 Passive Filters: Definition, types- Low pass filter (LPF), high pass filter (HPF), band pass filter(BPF) and band stop filter (BSF), features, frequency response, applications, comparison	Lecture Using Chalk-Board Video Demonstrations Assignments

CIRCUITS & NETWORKS**Course Code : 313325**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Analyze the phasor diagram of series A.C. circuit. TLO 4.2 Calculate active, reactive, apparent power and power factor for the specified circuit. TLO 4.3 Analyze the phasor diagram of parallel A.C. circuit. TLO 4.4 Determine admittance, conductance and susceptance for the given circuit. TLO 4.5 Interpret the output of the given R, L and C component at initial and final condition.	Unit - IV Single Phase A.C. Circuit 4.1 AC circuit with series element: R-L, R-C and R-L-C circuits, voltage and current waveform, impedance, reactance, phasor diagram, impedance triangle 4.2 Different types of Power: power factor, active (real) power, apparent power, reactive power, power triangle, power triangle of series AC circuit 4.3 AC circuit with parallel element: Resistance in parallel with pure inductance and capacitance and for series combination of resistance and inductance in parallel with capacitance, voltage and current waveform, impedance, reactance, phasor diagram, impedance triangle, power triangle for above circuit 4.4 Definition of admittance, conductance and susceptance 4.5 Initial and final conditions in switching circuits, meaning of $t=0^-$, $t=0^+$, $t=\infty$, R, L and C at initial and final conditions	Lecture Using Chalk-Board Video Demonstrations Assignment

CIRCUITS & NETWORKS**Course Code : 313325**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
5	TLO 5.1 Find the resonance condition for the specified series RLC circuit and calculate current, voltage, bandwidth, quality factor at resonant condition. TLO 5.2 Interpret the behaviour of series circuit with change in frequency. TLO 5.3 Find the resonance condition for the specified parallel RLC circuit and calculate current, voltage, bandwidth, quality factor at resonant condition. TLO 5.4 Interpret the behaviour of parallel RLC circuit with change input frequency. TLO 5.5 Describe the procedure to tune the given electrical circuit to achieve the resonance in the circuit.	Unit - V Resonance in Series and Parallel circuits 5.1 Resonance in series circuit: Impedance, phase angle, voltage, current, bandwidth, Quality factor (Q), magnification factor for series resonance circuit 5.2 Behaviour of RLC series circuit with change in input frequency 5.3 Resonance in Parallel Circuit: Impedance, phase angle, voltage, current, bandwidth, Quality factor (Q), magnification factor for parallel resonance circuit 5.4 Behaviour of RLC parallel circuit with change in input frequency	Lecture Using Chalk-Board Blended learning tools Assignments

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
---	--------------	---	-----------------------	---------------------

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

CIRCUITS & NETWORKS**Course Code : 313325**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Identify the loops in the given circuit. LLO 1.2 Verify KVL to find out the voltage across the element.	1	* Measure the voltage across resistive circuit and verify it, using Kirchhoff's Voltage law (KVL).	2	CO1
LLO 2.1 Identify the nodes in the given circuit. LLO 2.2 Verify KCL at given node.	2	Measure current in various branches of the given circuit and verify it, using Kirchhoff's current law (KCL).	2	CO1
LLO 3.1 Identify the meshes in given circuit. LLO 3.2 Use mesh analysis to calculate current through given branch.	3	* Measure current through and voltage across given branch of electric network and verify it by mesh analysis.	2	CO1
LLO 4.1 Identify the nodes in the given circuit. LLO 4.2 Use nodal analysis to calculate node voltage.	4	* Measure voltage at particular node and current through branch of network and verify it by nodal analysis.	2	CO1
LLO 5.1 Measure the current of the RL series circuit. LLO 5.2 Plot and interpret the transient response of given circuit on graph paper.	5	Observe transient response of RL series circuit with DC excitation.	2	CO1

CIRCUITS & NETWORKS**Course Code : 313325**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 6.1 Measure the voltage across capacitor in RC series circuit. LLO 6.2 Plot and interpret transient response of given circuit on graph paper.	6	* Observe transient response of RC series circuit with DC excitation.	2	CO1
LLO 7.1 Measure voltage and current of the given circuit. LLO 7.2 Verify Superposition theorem.	7	* Measure current through given branch of network and voltage across given element of the circuit and verify it applying Superposition theorem.	2	CO2
LLO 8.1 Measure load current of the given circuit. LLO 8.2 Verify Norton's theorem.	8	Measure short circuit current and Norton's resistance of the given circuit and verify it using Norton's theorem.	2	CO2
LLO 9.1 Measure load current of the given circuit. LLO 9.2 Verify Thevenin's theorem.	9	* Measure open circuit voltage and thevenin's resistance of the given circuit and verify it using Thevenin's theorem.	2	CO2
LLO 10.1 Verify Maximum power transfer theorem and calculate current ,voltage and power.	10	* Vary load resistance to transfer Maximum power in the given circuit using maximum power transfer theorem.	2	CO2

CIRCUITS & NETWORKS**Course Code : 313325**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 11.1 Verify the concept of interchangeability of sources and detectors in the given circuit.	11	Measure voltage to current ratio before and after interchanging the position of voltage source and current in the given circuit to verify reciprocity theorem.	2	CO2
LLO 12.1 Calculate input and output impedances of given network. LLO 12.2 Interpret the Z-parameters matrix.	12	* Measure input and output voltages and currents of the given two port network and calculate open circuit(Z) parameters for the given circuit.	2	CO3
LLO 13.1 Calculate Y parameters of given network. LLO 13.2 Interpret the Y parameters matrix.	13	Measure input and output voltages and currents of the given two port network and calculate short circuit(Y) parameters for given circuit.	2	CO3
LLO 14.1 Calculate ABCD parameters of given network. LLO 14.2 Interpret the ABCD parameters matrix.	14	Measure input and output voltages and currents of the given two port network calculate transmission(ABCD) parameters for given circuit.	2	CO3
LLO 15.1 Construct low pass filter using R and C and interpret the frequency response of RC Low Pass Filter.	15	Develop RC low pass filter on breadboard and plot its frequency response.	2	CO3

CIRCUITS & NETWORKS**Course Code : 313325**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 16.1 Construct high pass using R and C and interpret the frequency response of RC high pass filter.	16	* Develop RC high pass filter on breadboard and plot its frequency response.	2	CO3
LLO 17.1 Construct band pass filter using R and C and interpret the frequency response of RC band pass filter.	17	Develop RC band pass filter on breadboard and plot it's frequency response.	2	CO3
LLO 18.1 Construct symmetrical T attenuator. LLO 18.2 Interpret I/O of symmetrical T type attenuator.	18	* Test the performance of Symmetrical T attenuator.	2	CO3
LLO 19.1 Construct symmetrical Pi attenuator. LLO 19.2 Interpret I/O of PI type attenuator.	19	Test the performance of Symmetrical Pi attenuator.	2	CO3

CIRCUITS & NETWORKS**Course Code : 313325**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 20.1 Connect the R and L in series with A.C. supply and measure current and voltage across the circuit element. LLO 20.2 Interpret the phasor diagram of given RL series circuit for various input A.C. supply.	20	* Measure voltage and current in the given R-L series circuit and calculate active, reactive and apparent power consumed in the circuit.	2	CO4
LLO 21.1 Connect the R and C in series with A. C. supply. LLO 21.2 Interpret the phasor diagram of given RC series circuit for various input A.C. supply.	21	Measure voltage and current in the given R-C series circuit and calculate active, reactive and apparent power consumed in the circuit.	2	CO4
LLO 22.1 Connect the R, L and C in series with supply. LLO 22.2 Interpret the phasor diagram of given RLC circuit for various input A.C. supply	22	* Measure voltage and current in the given R-L-C series circuit and calculate active, reactive and apparent power consumed in the circuit.	2	CO4

CIRCUITS & NETWORKS**Course Code : 313325**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 23.1 Connect the R and C in parallel with supply. LLO 23.2 Interpret the phasor diagram of given RC parallel circuit for various input A.C. supply.	23	*Measure voltage and current in the given R-C parallel circuit and calculate power factor, active, reactive and apparent power consumed in the circuit.	2	CO4
LLO 24.1 Connect the R, L and C in parallel with supply. LLO 24.2 Interpret the phasor diagram of given R-L-C parallel circuit for various input A.C. supply.	24	Measure voltage and current in the given R-L-C parallel circuit and calculate power factor, active, reactive and apparent power consumed in the circuit.	2	CO4
LLO 25.1 Connect series connection of resistor and inductor in parallel with capacitor. LLO 25.2 Interpret the phasor diagram of given RL series circuit in parallel with C, for various input A.C. supply.	25	* Measure voltage and current in the given R-L-C parallel circuit consists of series connection of resistor and inductor in parallel with capacitor and calculate power factor, active, reactive and apparent power consumed in the circuit.	2	CO4
LLO 26.1 Measure and interpret initial and final condition of the capacitor in the given DC circuit.	26	Measure initial and final voltage across the capacitor before and after swicthing input supply.	2	CO4

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

CIRCUITS & NETWORKS**Course Code : 313325**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 27.1 Measure and interpret initial and final condition of the Inductor in the given DC circuit.	27	Measure initial and final current flowing through the inductive coil before and after switching the supply.	2	CO4
LLO 28.1 Tune the supply frequency to create resonance in given RLC series circuit.	28	* Measure voltage and current in the given RLC series circuit and calculate resonance frequency and impedance at resonance using variable supply frequency.	2	CO5
LLO 29.1 Tune the circuit parameters (L or C) and measure the resonance frequency of RLC series circuit .	29	Measure voltage and current in the given RLC series circuit and calculate resonance frequency and impedance at resonance by varying L or C.	2	CO5
LLO 30.1 Tune the supply frequency to create resonance in given RLC parallel circuit.	30	* Measure current of given RLC parallel circuit and calculate resonance frequency and impedance at resonance by varying supply frequency.	2	CO5
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

CIRCUITS & NETWORKS

Course Code : 313325

Micro project

- Prepare a report on real life applications of resonance (e.g. musical instruments)
- Prepare power point presentation on source transformation, star- delta transformation, mesh and nodal analysis and give presentation in the class.
- Build a circuit on breadboard with multiple resistors connected in series and measure voltage across each resistor. Verify using KVL.
- Prepare a chart for comparison of single phase series RLC and parallel RLC circuit and draw input and output waveforms also.
- Build a circuit to change brightness of lamp/change the speed of fan/ change the temperature of heater using basic components.
- Build a circuit on breadboard with multiple resistors connected in parallel and measure current across each resistor. Verify using KCL.

Assignment

- Calculate the Z, Y and ABCD parameters for the given two port network.
- Find circuit parameters of Single Phase AC series(R-L, R-C,R-L-C) and parallel (R-L, R-C,R-L-C) circuit, also draw its phasor diagram.
- Find the resonance condition for the specified series and RLC circuit and calculate current, voltage, bandwidth, quality factor .Observe the behaviour of R, L and C with change in frequency for series circuit.
- Identify the number of loops and nodes in the given circuit and solve the circuit using Nodal analysis and Mesh analysis.
- Simplify complex circuit using Thevenin's theorem, Norton's theorem and draw equivalent circuit for given circuit.

Activities For Specific Learning / Skills Development

- Verification of various network analysis and theorems in Virtual Laboratory (<https://asnm-iitkgp.vlabs.ac.in/>).
- Verification of various network analysis and theorems using Simulation Software (MATLAB, PSPICE).

CIRCUITS & NETWORKS**Course Code : 313325**

- Perform R-L-C circuit analysis in Virtual Laboratory (<https://asnm-iitkgp.vlabs.ac.in/exp/rlc-circuit-analysis/>).
- Experimental verification of frequency response of R-L-C series Circuit (<https://asnm-iitkgp.vlabs.ac.in/exp/rlc-series-circuit/>).
- Test the resonance in Series RLC circuit using Simulation Software (MATLAB, PSPICE).

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	Voltmeter PMMC Type: DC, 0-150/300 V, 0-250/500 V, 0-75/150 V	11,12,13,14,16
2	Ammeter PMMC Type: DC, 0-1.5/3 Amp, 0-2.5/5 Amp, 0-5/10 Amp	12
3	Single phase Autotransformer 0-270 V,50Hz,10 A	20,21,22,23,24,25
4	Digital Storage Oscilloscope: 2 and 4 analog channel models 100 and 70 MHz bandwidth models Up to 1 GS/s sampling rate.	26,27,28

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

CIRCUITS & NETWORKS**Course Code : 313325**

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
5	Signal Generator, Frequency : 0.1Hz ~ 5MHz ; Output waveforms : Sine, triangle, square, positive and negative pulse	26,27,28
6	Decade capacitor Box with Decades of 1nF, 10nF, 100nF, 1 μ F, 10 μ F	28,29,30
7	Decade Resistance Box with range 1 ohm to 1000 M ohm	28,29,30
8	Decade Inductor Box with Decades of 10 μ H, 100 μ H, 1mH, 10mH, 100mH	28,29,30
9	Ammeters MI type: AC/DC, 0-1 Amp, 0-1.5 Amp, 0-2.5 Amp, 0-5 Amp	All
10	Voltmeter MI Type: AC/DC, 0-150/300 V, 0-250/500 V, 0-75/150 V	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	DC Network Analysis	CO1	8	4	4	6	14
2	II	Network Theorems	CO2	12	2	6	10	18
3	III	Analysis of two port network	CO3	8	2	4	6	12
4	IV	Single Phase A.C. Circuit	CO4	11	4	4	6	14
5	V	Resonance in Series and Parallel circuits	CO5	6	2	4	6	12
Grand Total				45	14	22	34	70

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- Two offline unit tests of 30 marks and average of two-unit test marks will be consider for out of 30 marks.

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

CIRCUITS & NETWORKS**Course Code : 313325**

- Each practical will be assessed considering 60% weightage to process, 40% weightage to product.
- For formative assessment of laboratory learning 25 marks

Summative Assessment (Assessment of Learning)

- End semester summative assessment is of 25 marks for laboratory learning.
- End semester assessment is of 70 marks.

XI. SUGGESTED COS - POS MATRIX FORM

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2	3	2	1	1	1	1			
CO2	3	3	2	1	1	1	1			
CO3	2	2	3	1	1	1	1			
CO4	2	2	3	1	-	1	-			
CO5	2	1	2	1	-	1	-			

CIRCUITS & NETWORKS**Course Code : 313325**

Legends :- High:03, Medium:02,Low:01, No Mapping: -
 *PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Mittal V.N., Mittal Arvind	Basic Electrical Engineering	McGraw Hill Education, Noida,2005,ISBN: 978007093572
2	Paranjothi S.R.	Electric Circuit Analysis	New Age Publisher, New Delhi , 2011, ISBN:978-81-224-3154-4
3	Theraja B.L.,Theraja A.K.	A Text book of Electrical Technology Vol-I	S. Chand and Co. New Delhi, 2006 ISBN:978-81-219-2440-5
4	Boylested R.L.	Introductory Circuit Analysis	Wheeler New Delhi, 2013,ISBN: 978-0023131615
5	P. Ramesh Babu	Electric Circuit Analysis	Scitech Publication (India) Pvt. Ltd ISBN : 978 81 8371 078 7
6	Ravish R Sihng	Network Analysis and Synthesis	Mc Graw Hill Education (India) Pvt. Ltd. ISBN: 978-1-25-906295-7
7	Richard C. Doorf, James A. Svaboda	Introduction to electric Circuit	Wiley India Pvt. Ltd. ISBN: 978-81-265-5344-0
8	Sudhakar, A., Palli Shyammohan, S	Circuit and network	McGraw Hill, New Delhi, 2006 ISBN: 978-0-07-340458-5

XIII. LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	www.scilab.org/scilab	Open source simulator for simulation of theorems

CIRCUITS & NETWORKS**Course Code : 313325**

Sr.No	Link / Portal	Description
2	www.ni.com/multisim	Open source simulator for simulation of theorems and circuit analysis
3	https://www.nptelvideos.com/course.php?id=462	NPTEL Circuit Theory Video Lectures
4	https://asnm-iitkgp.vlabs.ac.in/	Virtual laboratory link for theorems, R-L-C circuit analysis and its frequency response
5	https://www.udemy.com/course/electrical-circuit-for-electric al-electronics-engineering/	Basics, circuit element, circuit solving, network theorems, transient analysis
6	https://everycircuit.com/app	Online and mobile app to design and simulate electronic circuits
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

BASIC PYTHON PROGRAMMING

Course Code : 313011

Programme Name/s	: Automation and Robotics/ Digital Electronics/ Electrical Engineering/ Electronics & Tele-communication Engg./ Electrical and Electronics Engineering/ Electrical Power System/ Electronics & Communication Engg./ Electronics Engineering/ Instrumentation & Control/ Industrial Electronics/ Instrumentation/ Medical Laboratory Technology/ Medical Electronics
Programme Code	: AO/ DE/ EE/ EJ/ EK/ EP/ ET/ EX/ IC/ IE/ IS/ ML/ MU
Semester	: Third / Fourth / Sixth
Course Title	: BASIC PYTHON PROGRAMMING
Course Code	: 313011

I. RATIONALE

Electronics based industries needs to deal with creating circuits design, simulation, signal processing and control systems which can be developed using Python. This course deals with the basics of python to enhance the programming skills of diploma students. The course will enable students to write python programs as well as use different python libraries to solve given problems.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to attain the following industry/employer expected outcome through various teaching learning experiences:

Develop programs using python to solve wide-reaching electronics engineering related problems.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

BASIC PYTHON PROGRAMMING**Course Code : 313011**

- CO1 - Develop script to demonstrate use of basic building blocks of python.
- CO2 - Implement conditional and looping statements for given problem statement.
- CO3 - Perform operations on sequence structures in python.
- CO4 - Implement basics of object oriented programming concepts.
- CO5 - Create modules and packages for given purpose.

IV. TEACHING-LEARNING & ASSESSMENT SCHEME

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL					FA-TH	SA-TH	Total		FA-PR		SA-PR		SLA		
Max	Max	Max	Min	Max	Min	Max	Min	Max	Min												
313011	BASIC PYTHON PROGRAMMING	BPP	AEC	2	-	2	-	4	2	-	-	-	-	-	25	10	25@	10	-	-	50

BASIC PYTHON PROGRAMMING**Course Code : 313011****Total IKS Hrs for Sem. : 0 Hrs**

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
-------	---	---	--------------------------------

BASIC PYTHON PROGRAMMING**Course Code : 313011**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Describe the given Keywords and Constants in Python. TLO 1.2 Use indentation, comments in the given program. TLO 1.3 Use different types of operators for writing expressions. TLO 1.4 Write python program using input output statements.	Unit - I Basic Python's Constructs 1.1 Introduction to Python- Python as scripting Language, Programming language Vs Scripting Language (C vs Python), Python's Technical Strength, Application in different domains 1.2 Python's building blocks- Identifiers, Keywords, Variables, Constants, Indentation, Comments in python 1.3 Python's Data Types – Numbers, Strings, List, Tuples, Dictionaries, Sets 1.4 Input and Output statements in python 1.5 Operators in Python- Operators as Arithmetic, Assignment, Unary Minus, Relational, Logical, Boolean, Bitwise, Membership, Identity, Operator precedence and Associativity	Presentations, Lecture Using Chalk-Board, Hands-on
2	TLO 2.1 Develop programs using Conditional Statements. TLO 2.2 Develop programs using Loop statements. TLO 2.3 Use statements to control the loops.	Unit - II Control Statements in Python 2.1 Types of Control Statements – Decision making statements, Looping statements 2.2 Decision Making Statements: - if, if...else, else-if ladder, nested if and switch statement 2.3 Looping statement: - while loop, for loop, nested loop 2.4 Manipulating Loops- use of break, continue and pass statements	Lecture Using Chalk-Board, Demonstration, Hands-on, Flipped Classroom

BASIC PYTHON PROGRAMMING**Course Code : 313011**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Develop program to manipulate List for given purpose. TLO 3.2 Develop program to manipulate Tuples for given purpose. TLO 3.3 Develop program to manipulate Sets for given purpose. TLO 3.4 Develop program to manipulate Dictionaries for given purpose.	Unit - III Data Structures in Python 3.1 List- Defining List, Creating list, Accessing values from list, Updating the elements of a list, Concatenation of two lists, Repeating of Lists, Membership in list, Aliasing and cloning Lists, Methods to process Lists, Nested Lists 3.2 Tuples- Defining Tuple, Creating Tuples, Accessing the Tuple elements, Inserting elements in a Tuple, modifying elements of a Tuple, Deleting elements from a Tuple, Basic operations in Tuples, Functions to process Tuples, Nested Tuples 3.3 Sets- Defining Set, Creating a Set, Accessing elements from set, Add and update Set, Remove an elements from a Set, Built in functions with Set, Set methods to perform mathematical operations, other relevant set methods 3.4 Dictionaries- Defining Dictionary, Creating Dictionary, Accessing elements from Dictionary, Add and update Dictionary, Delete an element from a Dictionary, Built in functions of Dictionary, Methods to perform Dictionary	Demonstration, Lecture Using Chalk-Board, Hands-on

BASIC PYTHON PROGRAMMING**Course Code : 313011**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Use python built-in functions to perform tasks. TLO 4.2 Develop relevant user defined function for the given purpose. TLO 4.3 Define classes to create and access objects with its methods and attributes.	Unit - IV Functions with Basic OOP concepts 4.1 Python Functions- Use of python built in functions (e.g. type/data conversion functions, math and string functions), User defined function- Function definition, function calling, function arguments and parameter passing, Return statement, scope of variables (Global and Local Variables) 4.2 Basic OOP concepts- Introduction to object-oriented programming, Creating classes and objects, Constructors and Destructors in python, Data abstraction and Encapsulation	Demonstration, Lecture Using Chalk-Board, Hands-on
5	TLO 5.1 Develop a python module in python for given purpose. TLO 5.2 Develop a python package for given purpose. TLO 5.3 Use NumPy for performing mathematical operations on arrays. TLO 5.4 Use matplotlib to create data visualization in python.	Unit - V Modules and Packages in Python 5.1 Modules- Writing modules, importing module, python built in modules (Numeric and mathematical module, Functional Programming Module) 5.2 Python packages- Introduction, Writing python packages, using standard packages (NumPy, matplotlib) and user defined package statements	Demonstration, Lecture Using Chalk-Board, Hands-on

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES.

BASIC PYTHON PROGRAMMING**Course Code : 313011**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 1.1 Install Python Integrated Development Environment.	1	a) Install and configure Python IDE. b) Write Python program to display message on screen.	2	CO1
LLO 2.1 Use operators in Python.	2	*a) Write simple Python program to calculate equivalent registers connected in series and parallel. Accept values of R1, R2 and R3 from the user. *b) Write simple Python program to calculate value of voltage by applying Ohm's law. Accept value of Current(I) and Resistance(R) from the user.	2	CO1
LLO 3.1 Implement two-way branching statement.	3	Write program to check whether entered frequency is radio frequency or audio frequency.	2	CO2
LLO 4.1 Implement multi-way branching statement.	4	*a) Write program to display various radio frequency bands using if..elseif ladder. *b) Write program to display resistor color code using switch statement.	2	CO2
LLO 5.1 Implement control loops for solving iterative problems.	5	*a. Write a simple Python program to demonstrate use of control loops: i) while ii) do while *b. Create a simple program, to demonstrate use of: for loop in Python (e.g.: various pattern building, printing multiplication table, checking palindrome number etc.)	2	CO2

MSBTE Approval Dt. 02/07/2024**Semester - 3 / 4 / 6, K Scheme**

BASIC PYTHON PROGRAMMING**Course Code : 313011**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 6.1 Perform basic operations on the Lists.	6	*Write Python program to perform following operations on List: a) Create b) Access c) Update d) Delete elements from list.	2	CO3
LLO 7.1 Execute various tuple operations.	7	Develop Python program to perform following operations on Tuples: a) Create b) Access c) Update d) Delete Tuple elements	2	CO3
LLO 8.1 Implement various set operations.	8	Write Python program to perform following operations on Set: a) Create b) Access c) Update d) Delete Access Set elements	2	CO3

BASIC PYTHON PROGRAMMING**Course Code : 313011**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 9.1 Execute various operations on Dictionaries.	9	*Create a program to perform following operations on Dictionaries in Python: a) Create b) Access c) Update d) Delete e) Looping through Dictionary	2	CO3
LLO 10.1 Use built-in mathematical functions and string functions in python.	10	a) *Create python program to demonstrate use of math built-in function. b) *Create python program to demonstrate use of string built-in function.	2	CO4
LLO 11.1 Create user defined functions in Python.	11	Write python programs to define function with arguments. a) Calculate factorial of a number b) Swapping of two variables	2	CO4
LLO 12.1 Implement function with default arguments.	12	Write programs to define function with default arguments.	2	CO4
LLO 13.1 Use built-in python mathematical modules.	13	*Create a program to demonstrate use of: Built-in module (e.g. numeric, mathematical functional and programming module) in Python.	2	CO5
LLO 14.1 Write user-defined module in python.	14	Write program to create a user-defined module (e.g.: building calculator) in python.	2	CO5
LLO 15.1 Use python built-in packages.	15	*Develop Python program to demonstrate use of NumPy package for creating, accessing and performing different array operations.	2	CO5

MSBTE Approval Dt. 02/07/2024**Semester - 3 / 4 / 6, K Scheme**

BASIC PYTHON PROGRAMMING**Course Code : 313011**

Practical / Tutorial / Laboratory Learning Outcome (LLO)	Sr No	Laboratory Experiment / Practical Titles / Tutorial Titles	Number of hrs.	Relevant COs
LLO 16.1 Implement user-defined packages in python.	16	Write program to demonstrate the use of user-defined packages in Python.	2	CO5
Note : Out of above suggestive LLOs - • '*' Marked Practicals (LLOs) Are mandatory. • Minimum 80% of above list of lab experiment are to be performed. • Judicial mix of LLOs are to be performed to achieve desired outcomes.				

VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)

Micro project

- Not Applicable

BASIC PYTHON PROGRAMMING**Course Code : 313011****Note :**

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED

Sr.No	Equipment Name with Broad Specifications	Relevant LLO Number
1	a) Computer System with all necessary peripherals and internet connectivity. b) Any relevant python IDE like IDLE/PyCharm/VSCode/Jupyter Notebook/Online Python Compiler.	All

IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Basic Python's Constructs	CO1	4	0	0	0	0
2	II	Control Statements in Python	CO2	4	0	0	0	0
3	III	Data Structures in Python	CO3	10	0	0	0	0

MSBTE Approval Dt. 02/07/2024**Semester - 3 / 4 / 6, K Scheme**

BASIC PYTHON PROGRAMMING**Course Code : 313011**

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
4	IV	Functions with Basic OOP concepts	CO4	6	0	0	0	0
5	V	Modules and Packages in Python	CO5	6	0	0	0	0
Grand Total				30	0	0	0	0

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- Each practical will be assessed considering – 60% weightage to process and – 40% weightage to product.

Summative Assessment (Assessment of Learning)

- End semester summative assessment of 25 marks for laboratory learning.

XI. SUGGESTED COS - POS MATRIX FORM

BASIC PYTHON PROGRAMMING**Course Code : 313011**

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	2						1			
CO2	2			1			2			
CO3	1	1	1	2			2			
CO4	1	2	2	2			2			
CO5	1	1	1	2			2			
Legends :- High:03, Medium:02,Low:01, No Mapping: - *PSOs are to be formulated at institute level										

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	Giancarlo Zaccone	Natural Computing with Python	BPB, ISBN:9789388511612
2	Martin C. Brown	Python: The Complete Reference	Tata McGraw Hill ISBN: 9789387572942
3	Yashwant Kanetkar	Let Us Python	BPB, ISBN: 978-9391392253
4	Kumar Naveen, Taneja Sheetal.	Python Programming: A modular approach	Pearson, ISBN: 978-9352861293

MSBTE Approval Dt. 02/07/2024**Semester - 3 / 4 / 6, K Scheme**

BASIC PYTHON PROGRAMMING**Course Code : 313011**

Sr.No	Author	Title	Publisher with ISBN Number
5	Mark Lutz and David Ascher	Learning Python	O'Reilly, ISBN: 978-1449355739
6	Paul Barry	Head First Python	O'Reilly, ISBN: 978-1449382674
7	John Guttag	Introduction to Computation and Programming Using Python	MIT Press, ISBN: 978-0262529624
8	David Beazley	Python Essential Reference	Addison-Wesley Professional, ISBN: 978-0672329784
9	Dr. R. Nageswara Rao	Core Python Programming	DREAMTECH PRESS, ISBN: 978-9386052308

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	https://www.programiz.com/python-programming	Python Programming
2	https://python-iitk.vlabs.ac.in/Introduction.html	Virtual Lab for Python Programming- Basic Constructs in Python
3	https://www.geeksforgeeks.org/python-programming-language/	Python Programming
4	https://intellipaat.com/academy/course/introduction-to-python-programming-free-course/	Online Course-Python Programming
5	https://www.w3schools.com/python/	Python Programming
6	https://www.tutorialspoint.com/python/index.htm	Python Programming
7	https://www.python.org/	Python Programming
8	https://spoken-tutorial.org/tutorial-search/?search_foss=Python+3.4.3&search_language=English	Spoken Tutorial on Python Programming

BASIC PYTHON PROGRAMMING**Course Code : 313011**

Sr.No	Link / Portal	Description
Note : Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students		

MSBTE Approval Dt. 02/07/2024**Semester - 3 / 4 / 6, K Scheme**

ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002****Programme Name/s**

**: Architecture Assistantship/ Automobile Engineering./ Artificial Intelligence/ Agricultural Engineering/
 Artificial Intelligence and Machine Learning/ Automation and Robotics/ Architecture/ Cloud Computing and Big Data/
 Civil Engineering/ Chemical Engineering/ Computer Technology/ Computer Engineering/
 Civil & Rural Engineering/ Construction Technology/ Computer Science & Engineering/
 Fashion & Clothing Technology/
 Dress Designing & Garment Manufacturing/ Digital Electronics/ Data Sciences/ Electrical Engineering/
 Electronics & Tele-communication Engg./ Electrical and Electronics Engineering/ Electrical Power System/ Electronics & Communication Engg./
 Electronics Engineering/ Food Technology/ Computer Hardware & Maintenance/ Hotel Management & Catering Technology/
 Instrumentation & Control/ Industrial Electronics/ Information Technology/ Computer Science & Information Technology/
 Instrumentation/ Interior Design & Decoration/ Interior Design/ Civil & Environmental Engineering/
 Mechanical Engineering/ Mechatronics/ Medical Laboratory Technology/ Medical Electronics/
 Production Engineering/ Printing Technology/ Polymer Technology/ Computer Science/ Textile Technology/ Electronics & Computer Engg./ Travel and Tourism/ Textile Manufactures/**

Programme Code

**: AA/ AE/ AI/ AL/ AN/ AO/ AT/ BD/ CE/ CH/ CM/ CO/ CR/ CS/ CW/ DC/ DD/ DE/
 DS/ EE/ EJ/ EK/ EP/ ET/ EX/ FC/ HA/ HM/ IC/ IE/ IF/ IH/ IS/ IX/ IZ/ LE/
 ME/ MK/ ML/ MU/ PG/ PN/ PO/ SE/ TC/ TE/ TR/ TX**

Semester**: Third****MSBTE Approval Dt. 02/07/2024****Semester - 3, K Scheme**

ESSENCE OF INDIAN CONSTITUTION

Course Code : 313002

Course Title : ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002****I. RATIONALE**

This course will focus on the basic structure and operative dimensions of Indian Constitution. It will explore various aspects of the Indian political and legal system from a historical perspective highlighting the various events that led to the making of the Indian Constitution. The Constitution of India is the supreme law of India. The document lays down the framework demarcating the fundamental political code, structure, procedures, powers, and sets out fundamental rights, directive principles, and the duties of citizens. The course on constitution of India highlights key features of Indian Constitution that makes the students a responsible citizen. In this online course, we shall make an effort to understand the history of our constitution, the Constituent Assembly, the drafting of the constitution, the preamble of the constitution that defines the destination that we want to reach through our constitution, the fundamental right constitution guarantees through the great rights revolution, the relationship between fundamental rights and fundamental duties, the futurist goals of the constitution as incorporated in directive principles and the relationship between fundamental rights and directive principles.

II. INDUSTRY / EMPLOYER EXPECTED OUTCOME

The aim of this course is to help the student to attain the following industry /employer expected outcome – Abide by the Constitution in their personal and professional life.

III. COURSE LEVEL LEARNING OUTCOMES (COS)

Students will be able to achieve & demonstrate the following COs on completion of course based learning

- CO1 - List salient features and characteristics of the constitution of India.
- CO2 - Follow fundamental rights and duties as responsible citizen of the country.
- CO3 - Analyze major constitutional amendments in the constitution.
- CO4 - Follow procedure to cast vote using voter-id.

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002****IV. TEACHING-LEARNING & ASSESSMENT SCHEME**

Course Code	Course Title	Abbr	Course Category/s	Learning Scheme					Credits	Assessment Scheme											
				Actual Contact Hrs./Week			SLH	NLH		Paper Duration	Theory				Based on LL & TL				Based on SL		Total Marks
															Practical						
				CL	TL	LL	FA-TH	SA-TH			Total		FA-PR		SA-PR		SLA				
Max	Max	Max	Min	Max	Min	Max	Min	Max	Min												
313002	ESSENCE OF INDIAN CONSTITUTION	EIC	VEC	1	-	-	1	2	1	-	-	-	-	-	-	-	-	-	50	20	50

Total IKS Hrs for Sem. : 0 Hrs

Abbreviations: CL- Classroom Learning , TL- Tutorial Learning, LL-Laboratory Learning, SLH-Self Learning Hours, NLH-Notional Learning Hours, FA - Formative Assessment, SA -Summative assessment, IKS - Indian Knowledge System, SLA - Self Learning Assessment

Legends: @ Internal Assessment, # External Assessment, *# On Line Examination , @\$ Internal Online Examination
Note :

1. FA-TH represents average of two class tests of 30 marks each conducted during the semester.
2. If candidate is not securing minimum passing marks in FA-PR of any course then the candidate shall be declared as "Detained" in that semester.
3. If candidate is not securing minimum passing marks in SLA of any course then the candidate shall be declared as fail and will have to repeat and resubmit SLA work.
4. Notional Learning hours for the semester are (CL+LL+TL+SL)hrs.* 15 Weeks
5. 1 credit is equivalent to 30 Notional hrs.
6. * Self learning hours shall not be reflected in the Time Table.
7. * Self learning includes micro project / assignment / other activities.

ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002****V. THEORY LEARNING OUTCOMES AND ALIGNED COURSE CONTENT**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
1	TLO 1.1 Explain the meaning of preamble of the constitution. TLO 1.2 Explain the doctrine of basic structure of the constitution. TLO 1.3 List the salient features of constitution. TLO 1.4 List the characteristics of constitution.	Unit - I Constitution and Preamble 1.1 Meaning of the constitution of India. 1.2 Historical perspectives of the Constitution of India. 1.3 Salient features and characteristics of the Constitution of India. 1.4 Preamble of the Constitution of India.	Presentations Blogs Hand-outs Modules Flipped classrooms Case studies
2	TLO 2.1 Enlist the fundamental rights. TLO 2.2 . Identify fundamental duties in general and in particular with engineering field. TLO 2.3 Identify situations where directive principles prevail over fundamental rights.	Unit - II Fundamental Rights and Directive Principles 2.1 Fundamental Rights under Part-III. 2.2 Fundamental duties and their significance under part-IV-A. 2.3 Relevance of Directive Principles of State Policy under part-IV A.	Presentations Blogs Hand-outs Modules Case Study Flipped Classroom

ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
3	TLO 3.1 Enlist the constitutional amendments. TLO 3.2 Elaborate the elements of Centre-State Relationship TLO 3.3 Analyze the purposes of various amendments.	Unit - III Governance and Amendments 3.1 3.1 Amendment procedure of the Constitution and their types - simple and special procedures. 3.2 The Principle of Federalism and its contemporary significance along with special committees that were setup. 3.3 Major Constitutional Amendment procedure - 1st, 7th, 42nd, 44th, 73rd & 74th, 76th, 86th, 52nd & 91st, 102nd	Cases of Federal disputes with relevant Supreme court powers and Judgements Presentations Blogs Hand-outs Problem based learning

ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002**

Sr.No	Theory Learning Outcomes (TLO's) aligned to CO's.	Learning content mapped with Theory Learning Outcomes (TLO's) and CO's.	Suggested Learning Pedagogies.
4	TLO 4.1 Explain the importance of electoral rights. TLO 4.2 Write the step by step procedure for process of registration TLO 4.3 Explain the significance of Ethical electoral participation TLO 4.4 Explain the steps to motivation and facilitation for electoral participation TLO 4.5 Enlist the features of the voter's guide TLO 4.6 Explain the role of empowered voter TLO 4.7 Write the steps of voting procedure TLO 4.8 Write steps to create voter awareness TLO 4.9 Fill the online voter registration form TLO TLO 4.10 Follow procedure to cast vote using voter-id.	Unit - IV Electoral Literacy and Voter's Education 4.1 Electoral rights , Electoral process of registration 4.2 Ethical electoral participation 4.3 Motivation and facilitation for electoral participation 4.4 Voter's guide 4.5 Prospective empowered voter 4.6 Voting procedure 4.7 Voter awareness 4.8 Voter online registration https://www.ceodelhi.gov.in/ELCdetails.aspx	Presentations Hand-outs Modules Blogs Problem based Learning

VI. LABORATORY LEARNING OUTCOME AND ALIGNED PRACTICAL / TUTORIAL EXPERIENCES : NOT**MSBTE Approval Dt. 02/07/2024****Semester - 3, K Scheme**

ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002**

APPLICABLE.**VII. SUGGESTED MICRO PROJECT / ASSIGNMENT/ ACTIVITIES FOR SPECIFIC LEARNING / SKILLS DEVELOPMENT (SELF LEARNING)****Assignment**

- Outline the procedure to submit application for Voter-id
- Assignments are to be provided by the course teacher in line with the targeted COs.

A1. Prepare an essay on Constitution of India .

A2 Prepare a comparative chart of Unique features of Indian Constitution of India and Constitution of USA

- Assignments are to be provided by the course teacher in line with the targeted COs. A1. Prepare an essay on Constitution of India . A2 Prepare a comparative chart of Unique features of Indian Constitution of India and Constitution of USA A3.

Self-learning topics: Parts of the constitution and a brief discussion of each part Right to education and girl enrollment in schools. GER of Girls and Boys. Right to equality. Social Democracy. Women Representation in Parliament and State Assemblies. LGBTQIA+

Micro project

- 1. Organize a workshop-cum discussions for spreading awareness regarding Fundamental Rights of the citizen of the country
- 2. Prepare elaborations where directive principle of State policy has prevailed over Fundamental rights with relevant Supreme Court Judgements.
- 3. Organize a debate on 42nd, 97th and 103rd Constitutional Amendment Acts of Constitution of India.

Seminar

ESSENCE OF INDIAN CONSTITUTION

Course Code : 313002

- 1 Differences in the ideals of Social democracy and Political democracy.
- 2 Democracy and Women's Political Participation in India.
- 3 Khap Panchayat - an unconstitutional institution infringing upon Constitutional ethos.
- 4 Situations where directive principles prevail over fundamental rights.

Group discussions on current print articles.

- - Art 356 and its working in Post-Independent India.
 - Women's Reservation in Panchayat leading to Pati Panchayats - Problems and Solutions.
 - Adoption of Article 365 in India.
 - Need of Amendments in the constitution.
 - Is India moving towards a Unitary State Model ?

Activity

- Arrange Mock Parliament debates.
Prepare collage/posters on current constitutional issues.
 - i. National (Art 352) & State Emergencies (Art 356) declared in India.
 - ii. Seven fundamental rights.
 - iii. Land Reforms and its effectiveness - Case study of West-Bengal and Kerala.

Cases: Suggestive cases for usage in teaching:

- A.K. Gopalan Case (1950) :SC contented that there was no violation of Fundamental Rights enshrined in Articles 13, 19, 21 and 22 under the provisions of the Preventive Detention Act, if the detention was as per the procedure established by law. Here, the SC took a narrow view of Article 21.
- Shankari Prasad Case (1951) : This case dealt with the amendability of Fundamental Rights (the First Amendment's validity was challenged). The SC contended that the Parliament's power to amend under Article 368 also includes the power to amend

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002**

the Fundamental Rights guaranteed in Part III of the Constitution.

Minerva Mills case (1980) : This case again strengthens the Basic Structure doctrine. The judgement struck down 2 changes made to the Constitution by the 42nd Amendment Act 1976, declaring them to violate the basic structure. The judgement makes it clear that the Constitution, and not the Parliament is supreme.

Maneka Gandhi case (1978) : A main issue in this case was whether the right to go abroad is a part of the Right to Personal Liberty under Article 21. The SC held that it is included in the Right to Personal Liberty. The SC also ruled that the mere existence of an enabling law was not enough to restrain personal liberty. Such a law must also be “just, fair and reasonable.”

Other cases:

1. Kesavananda Bharati Case (1973) : In this case the Hon. SC laid down a new doctrine of the ‘basic structure’ (or ‘basic features’) of the Constitution. It ruled that the constituent power of Parliament under Article 368 does not enable it to alter the ‘basic structure’ of the Constitution. This means that the Parliament cannot abridge or take away a Fundamental Right that forms a part of the ‘basic structure’ of the Constitution.

2. Mathura Rape Case (1979) : A tribal woman Mathura (aged 14 to 16 years) was raped in Police Custody. The case raised the questions on the idea of 'Modesty of Woman' and here it was a tribal woman who succumbs to multiple patriarchies. Custodial rape was made an offence and was culpable with the detainment of 7 years or more under Section 376 of Indian Penal Code. The weight of proofing the allegations moved from the victim to the offender, once sexual intercourse is established. The publication of the victim's identity was banned and it was also held that rape trials should be conducted under the cameras.

3. Puttsamy vs Union of India (2017) : In this landmark case which was finally pronounced by a 9-judge bench of the Supreme Court on 24th August 2017, upholding the fundamental right to privacy emanating from Article 21. The court stated that Right to Privacy is an inherent and integral part of Part III of the Constitution that guarantees fundamental rights. The conflict in this area mainly arises between an individual's right to privacy and the legitimate aim of the government to implement its policies and a balance needs to be maintained while doing the same.

4. Navtej Singh Johar & Ors. v. Union of India (2018) : Hon. SC Decriminalised all consensual sex among adults, including homosexual sex by scrapping down section 377 of the Indian penal code (IPC). The court ruled that LGBTQ community are equal citizens and underlined that there cannot be discrimination in law based on sexual orientation and gender.

5. Anuradha Bhasin Judgement (2020) : The Supreme Court of India ruled that an indefinite suspension of internet services would be illegal under Indian law and that orders for internet shutdown must satisfy the tests of necessity and proportionality.

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002**

The Court reiterated that freedom of expression online enjoyed Constitutional protection, but could be restricted in the name of national security. The Court held that though the Government was empowered to impose a complete internet shutdown, any order(s) imposing such restrictions had to be made public and was subject to judicial review.

Note :

- Above is just a suggestive list of microprojects and assignments; faculty must prepare their own bank of microprojects, assignments, and activities in a similar way.
- The faculty must allocate judicious mix of tasks, considering the weaknesses and / strengths of the student in acquiring the desired skills.
- If a microproject is assigned, it is expected to be completed as a group activity.
- SLA marks shall be awarded as per the continuous assessment record.
- For courses with no SLA component the list of suggestive microprojects / assignments/ activities are optional, faculty may encourage students to perform these tasks for enhanced learning experiences.
- If the course does not have associated SLA component, above suggestive listings is applicable to Tutorials and maybe considered for FA-PR evaluations.

VIII. LABORATORY EQUIPMENT / INSTRUMENTS / TOOLS / SOFTWARE REQUIRED : NOT APPLICABLE**IX. SUGGESTED WEIGHTAGE TO LEARNING EFFORTS & ASSESSMENT PURPOSE (Specification Table)**

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
1	I	Constitution and Preamble	CO1	4	0	0	0	0
2	II	Fundamental Rights and Directive Principles	CO2	4	0	0	0	0
3	III	Governance and Amendments	CO3	4	0	0	0	0

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002**

Sr.No	Unit	Unit Title	Aligned COs	Learning Hours	R-Level	U-Level	A-Level	Total Marks
4	IV	Electoral Literacy and Voter's Education	CO4	3	0	0	0	0
Grand Total				15	0	0	0	0

X. ASSESSMENT METHODOLOGIES/TOOLS**Formative assessment (Assessment for Learning)**

- Assignment, Self-learning and Terms work Seminar/Presentation

Summative Assessment (Assessment of Learning)**XI. SUGGESTED COS - POS MATRIX FORM**

Course Outcomes (COs)	Programme Outcomes (POs)							Programme Specific Outcomes* (PSOs)		
	PO-1 Basic and Discipline Specific Knowledge	PO-2 Problem Analysis	PO-3 Design/ Development of Solutions	PO-4 Engineering Tools	PO-5 Engineering Practices for Society, Sustainability and Environment	PO-6 Project Management	PO-7 Life Long Learning	PSO-1	PSO-2	PSO-3
CO1	1	-	-	-	2	-	-			

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002**

CO2	1	-	-	-	2	-	-			
CO3	1	2	-	-	2	-	1			
CO4	-	-	-	1	-	-	-			

Legends :- High:03, Medium:02,Low:01, No Mapping: -
 *PSOs are to be formulated at institute level

XII. SUGGESTED LEARNING MATERIALS / BOOKS

Sr.No	Author	Title	Publisher with ISBN Number
1	P.M.Bakshi	The Constitution of India	Universal Law Publishing, New Delhi 15th edition, 2018, ISBN: 9386515105 (Check the new edition)
2	D.D.Basu	Introduction to Indian Constitution	Lexis Nexis Publisher, New Delhi, 2015, ISBN:935143446X
3	B. K. Sharma	Introduction to Constitution of India	PHI, New Delhi, 6th edition, 2011, ISBN:8120344197

ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002**

Sr.No	Author	Title	Publisher with ISBN Number
4	MORE READS :	Oxford Short Introductions - The Indian Constitution by Madhav Khosla. The Indian Constitution: Cornerstone of a Nation by Granville Austin. Working a Democratic Constitution: A History by Garnville Austin Founding Mothers of the Indian Republic: Gender Politics of the Framing of the Constitution by Achyut Chetan. Our Parliament by Subhash C. Kashyap. Our Political System by Subhash C. Kashyap. Our Constitution by Subhash C. Kashyap. Indian Constitutional Law by Rumi Pal.	Extra Read
5	B.L. Fadia	The Constitution of India	Sahitya Bhawan, Agra, 2017, ISBN:8193413768

XIII . LEARNING WEBSITES & PORTALS

Sr.No	Link / Portal	Description
1	http://www.legislative.gov.in/constitution-of-india	Constitution overview
2	https://en.wikipedia.org/wiki/Constitution_of_India	Parts of constitution
3	https://www.india.gov.in/my-government/constitution-india	Constitution overview
4	https://www.toppr.com/guides/civics/the-indian-constitution/the-constitution-of-india/	Fundamental rights and duties
5	https://main.sci.gov.in/constitution	Directive principles
6	https://legalaffairs.gov.in/sites/default/files/chapter%203.pdf	Parts of constitution

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**

ESSENCE OF INDIAN CONSTITUTION**Course Code : 313002**

Sr.No	Link / Portal	Description
7	https://www.concourt.am/armenian/legal_resources/world_constitutions/constit/india/india-e.htm	Parts of constitution
8	https://constitutionnet.org/vl/item/basic-structure-indian-constitution	Parts of constitution

Note :

- Teachers are requested to check the creative common license status/financial implications of the suggested online educational resources before use by the students

MSBTE Approval Dt. 02/07/2024**Semester - 3, K Scheme**